Broad-scale Proterozoic deformation of the central Superior Province revealed by paleomagnetism of the 2.45 Ga Matachewan dyke swarm[1]

M. P. BATES[2] AND H. C. HALLS

Department of Geology, University of Toronto, Erindale Campus, Mississauga, Ont., Canada L5L 1C6

Received January 30, 1991
Revision accepted June 12, 1991

An extensive paleomagnetic study of the 2.45 Ga Matachewan dyke swarm of the North American Superior Province suggests that the interior of an Archean shield can undergo broad-scale distortion as a result of later (Proterozoic) orogenic activity around the craton margins. Data collected from over 300 sites, of which 137 are reported here for the first time, reveal that the dykes contain a dual-polarity primary remanence that varies across the swarm in both inclination and declination. These regional variations are statistically significant at the 95% confidence level, and cannot be attributed to remagnetization or to magnetic anisotropy. Inclination variation is probably due to real or apparent polar wander during the emplacement of the swarm, and may in part explain the declination variation as well. However, for dykes within and northwest of the Kapuskasing Structural Zone (KSZ) a positive correlation is found between regionally averaged values of declination and dyke trend. Here the dykes appear to have suffered differential rotations about vertical axes of up to 40° since emplacement. The Matachewan swarm radiates northwards from a broad focus situated approximately in northern Lake Huron but the trend of the western half of the swarm follows a broad Z-shaped pattern where it crosses the KSZ. Our data suggest that this changing trend is a secondary feature and that the western dykes, like their eastern counterparts, originally had a more uniform trend. This large-scale distortion of the western Matachewan swarm and Archean host rocks within and north of the KSZ is probably the result of broad-scale deformation during the Trans-Hudson Orogeny at about 1.95 Ga, coeval with uplift along the KSZ.

Une étude paléomagnétique couvrant une grande superficie de l'essaim de dykes de Matachewan, âgé de 2,45 Ga, dans la province du lac Supérieur de l'Amérique du Nord, révèle que l'intérieur du bouclier archéen peut déformer à grande échelle par des événements orogéniques subséquents (Protérozoïque) produits en bordure des marges du craton. Les données recueillies sur plus de 300 sites, dont 137 sont rapportées ici pour la première fois, indiquent que les dykes portent une rémanence primaire de polarité-double, dont l'inclinaison et la déclinaison varient au travers l'essaim. Ces variations régionales sont statistiquement significatives à un niveau de confiance de 95%, et elles ne peuvent être interprétées comme le résulat d'une réaimantation ou d'une anisotropie magnétique. La variation de l'inclinaison est probablement due à la dérive des pôles apparente ou réelle durant la période de mise en place de l'essaim, et peut en même temps expliquer une partie du changement de déclinaison. Cependant, pour les dykes localisés à l'intérieur et au nord-ouest de la zone structurale de Kapuskasing (ZSK), une corrélation positive est évidente entre les moyennes régionales des valeurs de déclinaison et la direction des dykes. Ici, les dykes semblent avoir été dérangés par des rotations différentielles autour d'axes verticaux, jusqu'à 40° depuis leur mise en place. L'essaim de dykes de Matachewan rayonne vers le nord à partir d'un foyer large qui est situé approximativement dans le secteur nord du lac Huron, mais la direction dans la partie occidentale de l'essaim suit une large structure en forme de Z en traversant la ZSK. Nos données suggèrent que cette déviation est une particularité secondaire et que les dykes situés à l'ouest, comme leurs contre-parties à l'est, avaient initialement une direction plus uniforme. Cette distortion à grande échelle de la partie occidentale de l'essaim de Matachewan et des roches encaissantes archéennes, à l'intérieur et au nord de la ZSK, reflète possiblement une déformation à grande échelle produite durant l'orogenèse du Trans-Hudsonien, il y a environ 1,95 Ga, et contemporaine du soulèvement survenu le long de la ZSK.

[Traduit par la rédaction]

Can. J. Earth Sci. **28**, 1780–1796 (1991)

Introduction

Much of the variation in paleomagnetic remanence from a given continent is due to local tectonics on a variety of scales. By comparing results from deformed rocks with stable, undisturbed areas, valuable insight into the nature of tectonic processes can be achieved. This technique has been successfully employed in orogenic belts along continental margins (e.g., Van der Voo and Channel 1989), and has led to an awareness of large-scale intracontinental deformation (e.g., Kissel and Laj 1989). The latter observation is of particular importance to paleomagnetists because an apparent polar wander path (APWP) typically includes data from continental shields assumed to have remained stable since the time of their original cratonization. In this paper we document the broad-scale deformation of the central Superior Province of North America during the early Proterozoic as revealed by the paleomagnetic signature of the giant Matachewan dyke swarm, and discuss the implications for the development of the Superior Province and the Laurentian APWP.

Background geology

The Superior Province is the largest of the Archean blocks that amalgamated during the Proterozoic to form the Laurentian craton (Hoffman 1988). It is composed of a number of supracrustal volcanic and sedimentary belts, deformed in the Kenoran orogeny (ca 2.6 Ga), alternating with more gneissic terrains. In the southern Superior Province where the Matachewan swarm is found, these belts trend in an east-northeast direction and define a number of subprovinces (Fig. 1). Recognized Proterozoic tectonic activity is restricted to orogens bordering the Superior Province, and to the Kapuskasing Structural Zone (KSZ) (Thurston *et al.* 1977), a north-northeast-trending discontinuous belt of granulite to upper amphibolite rocks thrust to the surface around 1.95 Ga ago (Percival and Card 1985; Percival *et al.* 1988; Fig. 1) along the west-dipping Ivanhoe Lake Cataclastic Zone (ILCZ) (Fig. 1; Cook 1985; Geis *et al.* 1990). The eastern

[1]Lithoprobe Phase II Publication 233; International Geological Correlation Program Project 257 on Precambrian dyke swarms, Publication 38.

[2]Present address: Department of Physics, University of Zimbabwe, P.O. Box MP 167, Mount Pleasant, Harare, Zimbabwe.

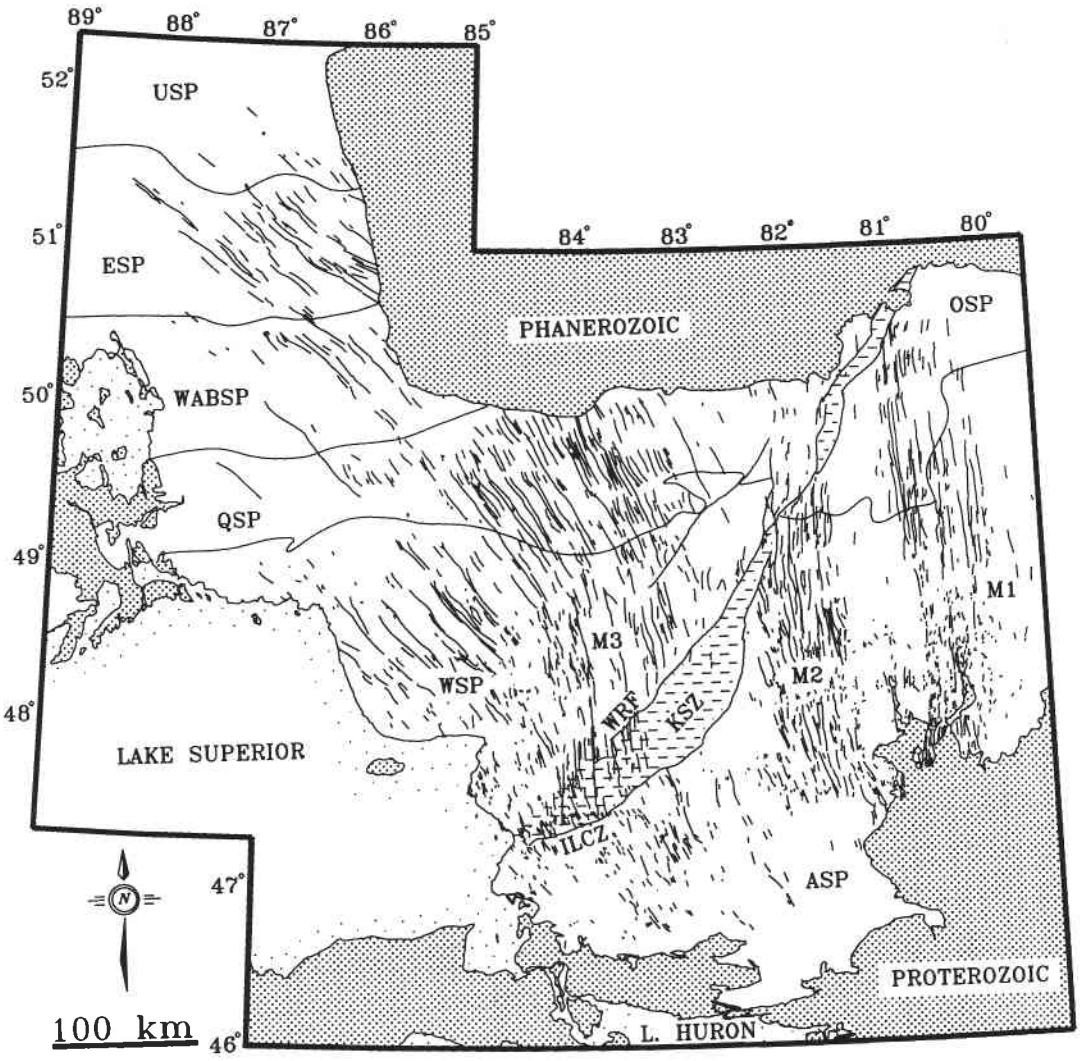


FIG. 1. Outline of the Matachewan dyke swarm in the central Superior Province. The location of the Kapuskasing Structural Zone is shown (KSZ, dashed pattern), bounded by the Ivanhoe Lake Cataclastic Zone (ILCZ) and the Wakusimi River Fault (WRF). The M1 and M2 subswarms are mainly east of the KSZ; the M3 subswarm describes an open Z-pattern as it traverses the KSZ. Subprovinces: ASP, Abitibi; WSP, Wawa; OSP, Opatica; QSP, Quetico; WABSP, Wabigoon; ESP, English River; USP, Uchi.

edge of the KSZ is marked by an abrupt change in metamorphic grade from granulite to greenschist facies across the ILCZ, but a more gradual change via amphibolite grade bounds the west side, representing a cross section through the middle crust (Percival 1983; Percival and Card 1985).

At the beginning of the Proterozoic, the central Superior Province was intruded by the Matachewan diabase dyke swarm (U–Pb age 2452^{+3}_{-2} Ma, Heaman 1989). The dykes radiate from a broad focal region in the south, possibly related to failed rifting (Fahrig 1987) or to plume activity (Halls and Bates 1990) and extend north and northwest for distances up to 700 km to form a 500 km wide fan (Fig. 1). The dykes are quartz tholeiites often with characteristic green feldspar phenocrysts up to 6 cm across (e.g., Condie *et al.* 1987; Phinney *et al.* 1988). They have suffered no more than a lower greenschist facies hydrous alteration, the development of which appears inversely related to the regional metamorphic grade of the host rocks, and reaches a maximum south of the Kapuskasing Zone (Halls and Palmer 1990).

In detail, the Matachewan dykes define three subswarms (M1–M3 from east to west, Fig. 1). M3 strikes northwest through the Abitibi Subprovince, traverses the KSZ with a northerly strike, and gently curves into a west-northwest trend in the western subprovinces, describing a broad Z-pattern (West and Ernst 1991). Although the dykes appear on both sides of the KSZ (Ernst and Halls 1984), they are not continuous across it (Bates and Halls 1990). Furthermore, within the southern KSZ the dykes appear to be absent from the highest grade granulite terrains, causing a gap in the swarm, but occur in amphibolite-grade rocks to the west (Halls *et al.* 1985). M1 and M2 strike north to north-northwest and are mainly restricted to the Abitibi and Opatica subprovinces, immediately east of the KSZ, although they impinge on and are disturbed by the KSZ at their northern extremities.

The dykes are on average within 5° of the vertical (Halls and Bates 1990), except around the eastern rim of the Lake Superior basin and in the southern KSZ where their average dip is about 80°E as a result of westward tilting and uplift (Card *et al.* 1981; Halls and Shaw 1988). Other geological evidence for post-Matachewan uplift within the southern KSZ is provided by amphibole geochemistry of the Matachewan dykes (Palmer and Barnett 1988), cloudy feldspar development in both Matachewan

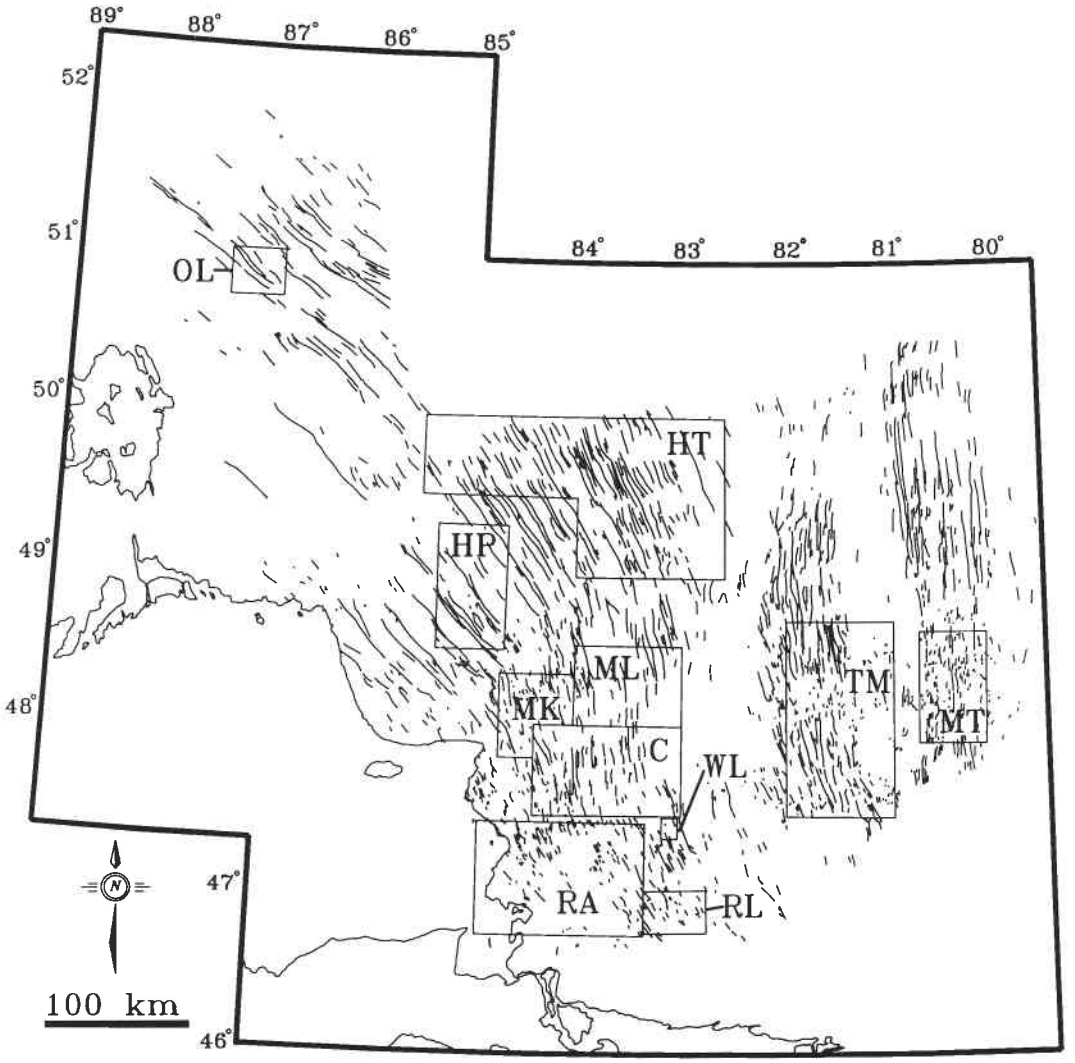


FIG. 2. Name and location of paleomagnetic data sets for the Matachewan dykes from this and previous work. OL, Ogoki Lake; HP, Hornepayne; HT, Hearst; MK, Manitowik Lake; and TM, Timmins (this paper, see Fig. 3); ML, Missinaibi Lake (Halls and Palmer 1990); C, Chapleau—Highway 101 (Halls and Palmer 1990); WL, Wenebegon Lake (Bates and Halls 1990); RL, Rocky Island Lake (Bates and Halls 1990); RA, Ranger Lake and Montreal River (Halls and Shaw 1988); MT, Matachewan (Irving and Naldrett 1977) and Otto stock (Buchan *et al.* 1990).

and 2.04 Ga Kapuskasing dykes (Halls and Palmer 1990), and by pseudotachylyte fracturing of Kapuskasing dykes adjacent to the ILCZ (Hanes *et al.* 1988).

Paleomagnetism of Matachewan dykes

The large areal extent and relatively pristine condition of the Matachewan swarm are conducive for using paleomagnetism to observe any tectonic events that have affected the central Superior Province since the end of the Archean. To this end, we have carried out paleomagnetic analysis on more than 2000 oriented samples from about 300 sites within the M2 and M3 subswarms, of which 137 are reported here for the first time. Although the full extent of the swarm is known from well-defined aeromagnetic anomalies (West and Ernst 1991), exposure is often poor, and most of the sites were collected from road cuts or lake shores. New paleomagnetic data have been gathered from the Timmins and Gogama areas (TM sites) in the M2 subswarm east of the KSZ, and from the Hornepayne, Hearst, Manitowik Lake, and Ogoki Lake areas (HP, HT, MK, and OL sites respectively) in the M3 subswarm west of the KSZ, which in conjunction with previously published data affords extensive coverage of the entire dyke swarm (Figs. 2 and 3).

Experimental procedure

Depending on the quality of outcrop, up to eight block samples were taken per dyke, concentrating on chilled margins, which tend to give the most stable paleomagnetic signature. Each block was oriented in the field using both the sun and magnetic compass, and subsequently drilled and cut into 2.4 × 2.4 cm cylindrical specimens in the laboratory. Individual specimens (usually one per sample) were subject to detailed stepwise alternating-field (AF) demagnetization using a Schonstedt TD-1 instrument and an averaging procedure (Halls 1986) at higher fields to reduce scatter. Remanence was measured on a modified DIGICO spinner magnetometer. Data were analyzed using principal-component analysis (Kirschvink 1980, maximum angular deviation $\leq 10°$), remagnetization circles (Bailey and Halls 1984), and orthogonal vector plots (Zijderveld 1967). Sample remanence directions were then averaged for each dyke to produce a site mean. Dykes located within a few metres of

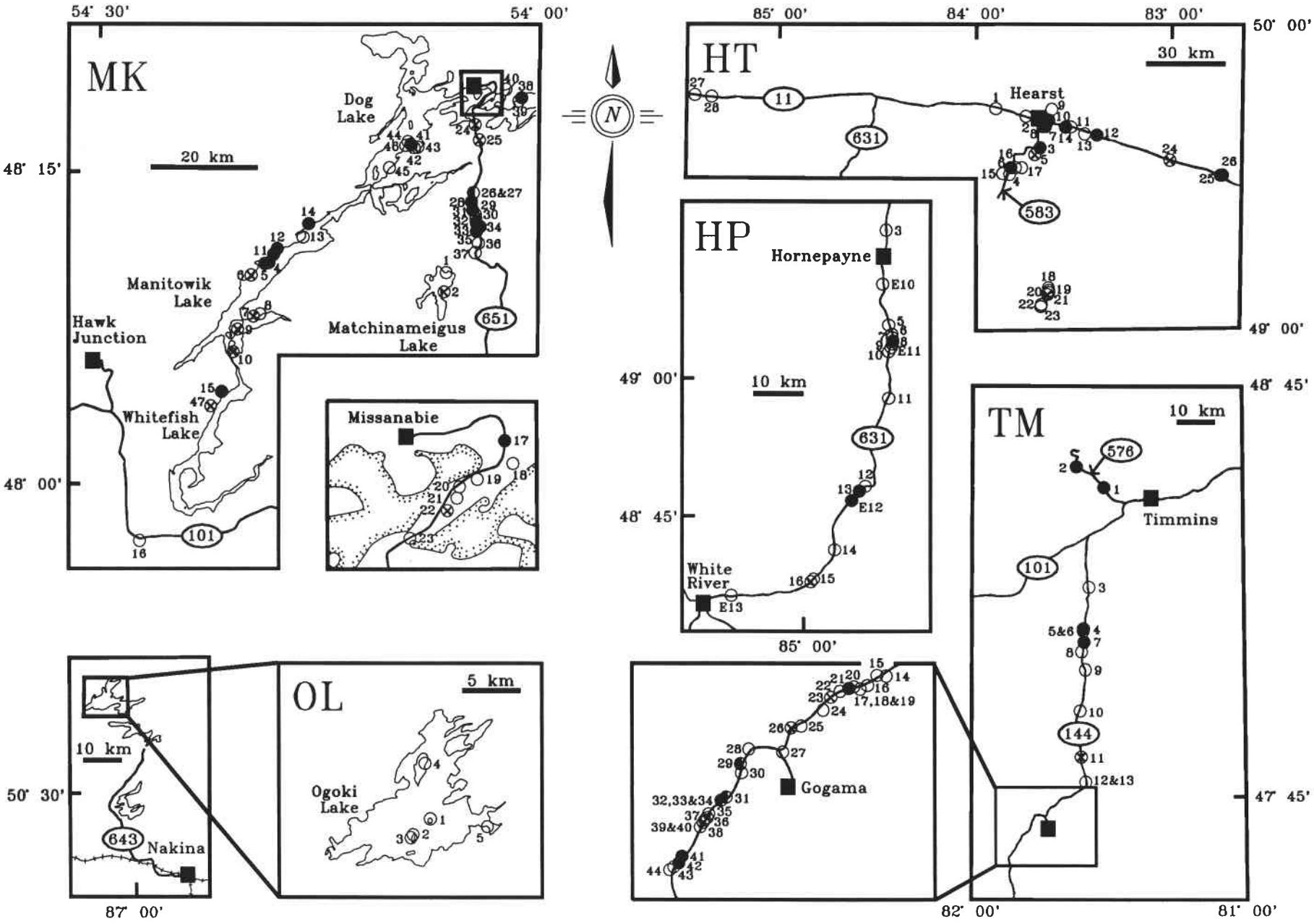


FIG. 3. Site locations for all new data presented in this paper. Letters in the corner of each map correspond with those on the location map in Fig. 2. HP, Hornepayne sites, plus some E sites from Ernst and Halls (1984); HT, Hearst; MK, Manitowik Lake; OL, Ogoki Lake; TM, Timmins. Solid (open) circles indicate N (R) polarity dykes. Dykes that failed to give a Matachewan remanence are marked by an X in a circle.

TABLE 1. Previously published paleomagnetic data for Matachewan dykes

Location[a]	Polarity[b]	N^c	Declination (°)	Inclination (°)	k^d	$\alpha_{95}{}^e$ (°)	$\Phi_{63}{}^f$ (°)	δ^g (°)	σ^g (°)
MT	N	—	—	—	—	—	—	360	—
	R	10	207.0	−16.0	54	7.0	—		
OS	N	—	—	—	—	—	—	360	—
	R	16	209.5	−14.8	39	6.0	—		
RA	N	3	23.2	17.9	49	17.9	8	326	9
	R	17	204.1	−11.6	52	5.4	10		
RL	N	3	19.2	20.7	45	18.6	9	325	16
	R	19	209.2	−6.0	42	5.2	9		
WL	N	1	15.7	24.9	—	—	—	350	13
	R	3	216.5	−9.7	190	9.1	4		
C and ML	N	37	30.1	22.4	66	2.9	—	—	—
	R	19	224.2	−23.4	41	5.3			

[a]MT, Matachewan (Irving and Naldrett 1977); OS, Otto stock (Buchan *et al.* 1990); RA, Ranger Lake, Montreal River (Halls and Shaw 1988); RL, WL, Rocky Island Lake, Wenebegon Lake (Bates and Halls 1990); C and ML, Chapleau—Highway 101, Missinaibi Lake (Halls and Palmer 1990). The data of Strangway (1964) and Fahrig *et al.* (1965) from the eastern part of the swarm were omitted because insufficient AC fields were employed (≤10 mT) to fully resolve the characteristic remanence. The data of Halls and Palmer (1990) supercede those reported by Ernst and Halls (1984). The mean values reported from Halls and Shaw (1988) include all site means regardless of quality except those from dykes affected by Keweenawan rifting along Lake Superior. Locations of studies are shown in Fig. 2. The C and ML data have been reevaluated and incorporated into the new data, and trend statistics can be found in Table 2.

[b]N, normal; R, reversed.

[c]N, number of sites.

[d]k, estimate of precision parameter.

[e]α_{95}, semiangle of the cone of 95% confidence.

[f]Φ_{63}, angular standard deviation of paleomagnetic data.

[g]δ, σ, mean, standard deviation of dyke trend; calculated once only for each data set using both N and R data (opposite polarity dykes in any one region show no significant difference in trend). The mean trend for the MT data is an estimate, and values of $\Phi_{63}$and σ are not available.

each other with the same paleomagnetic signature were considered to be branches of the same dyke, and results were combined to give a single site mean.

Previous results

The earliest paleomagnetic studies of the Matachewan dykes revealed a high-coercivity, high-unblocking-temperature, dual-polarity remanence with approximately antipodal directions near 030°/+20° and 210°/−20° (Strangway 1964; Fahrig *et al.* 1965), hereafter referred to as N (normal) and R (reverse). Most dykes carry only an N or an R remanence, with R in the majority. Since then, further studies have confirmed that this dual polarity remanence can be recovered from the entire swarm (Irving and Naldrett 1977; Aibangbee 1982; Ernst and Halls 1984; Halls and Shaw 1988; Halls and Palmer 1990; Bates and Halls 1990; Buchan *et al.* 1990) and is diagnostic of the Matachewan dykes. Previous results are summarized in Table 1.

Field tests show that both the N and R remanences are primary, and are probably variable combinations of thermal remanent and chemical remanent magnetization produced during original cooling of the dykes at 2.45 Ga. For both N and R dykes, the baked country rocks carry the same remanence as the dykes, but farther from the dyke margin an older remanence is recovered (Irving and Naldrett 1977; Schutts and Dunlop 1981; Buchan *et al.* 1990). Furthermore, dykes of opposite polarity exist in close proximity, and in two examples N dykes are found to crosscut R dykes (Halls 1991).

Although contact tests on C1 Matachewan dykes (see Table 2) in upper-amphibolite-grade KSZ gneisses are negative (H. C. Halls, unpublished data), a primary remanence is still possible because the KSZ has been uplifted after dyke intrusion (e.g., Hanes *et al.* 1988; Percival *et al.* 1988; Halls and Palmer 1990), so higher ambient temperatures at the time of intrusion would result in wide zones of remagnetization adjacent to the dykes.

Field and paleomagnetic observations at 14 sites involving crosscutting or juxtaposed dykes are consistent with there having been only a single reversal (R → N) during Matachewan time (Halls 1991). This conclusion is supported by U–Pb data on zircon and baddeleyite from N and R dykes which suggest that the Matachewan swarm was emplaced in a relatively short time span, perhaps less than 5 Ma (Heaman 1989). However, only three dykes have been studied and further sampling may reveal an igneous event of longer duration.

The tectonic significance of abrupt changes in polarity occurring across segments of the swarm is discussed elsewhere (Bates and Halls 1990; Halls and Bates 1990; Halls and Palmer 1990) but suffice to say that a change in dominant polarity from R to N along strike of the M3 subswarm across the southern KSZ boundary fault provides evidence of differential uplift across this fault in post-Matachewan time.

New paleomagnetic results

The intensity of natural remanent magnetization varies from less than 0.1 to more than 10 A/m, and after AF demagnetization the vast majority of samples analyzed is found to carry a characteristic Matachewan dyke remanence, isolated either as a stable end point or through converging remagnetization circles. Thermal demagnetization has been used in previous studies (Bates and Halls 1990), where the NRM is only unblocked after heating to greater than 500°C and then rapidly falls in intensity and becomes unstable at around 580°C. Although failing to separate magnetization components, the thermal treatment suggested that the remanence is carried by fine-grained single-domain magnetite (Pullaiah *et al.* 1975). Further details on the

TABLE 2. Paleomagnetic directions for different segments of the Matachewan swarm

Location	Polarity	N	Declination (°)	Inclination (°)	k	α_{95} (°)	Φ_{63} (°)	δ (°)	σ (°)
OL	R	4	185.4	−23.1	110	8.9	6	310	3
HP	N	3	10.3	24.0	220	8.4	4	326	10
	R	11	191.3	−25.6	100	4.5	6		
HT	N	6	14.3	31.6	89	7.1	6	337	10
	R	10	198.5	−17.8	66	6.0	7		
MK	N	15	38.9	32.3	40	6.1	9	344	10
	R	14	222.6	−23.8	24	8.2	12		
TM	N	5	21.4	28.2	430	3.7	3	347	12
	R	15	209.4	−14.8	87	4.1	6		
C1	N	14	26.4	23.8	87	4.3	6	353	10
C2	N	1	32.0	31.0	—	—	—	353	10
	R	5	228.9	−28.4	47	11.3	8		
C3	N	5	30.5	24.5	94	7.9	6	339	10

NOTES: Site level data used to calculate the means are tabulated in Table A1, Appendix, or if previously published, listed in Table 2A, Appendix. Locations refer to different segments of the swarm and include a combination of new (OL → TM) and previously published (C1 → C3) data (see Table A2, Appendix). Abbreviations as in Table 1.

nature of the remanence are given in Halls and Palmer (1990), Bates and Halls (1990), and Halls (1991); examples of sample AF demagnetization results from the new data set are shown on stereoplots and orthogonal vector plots in Fig. 4.

Site mean values from the six new study areas are given in Table A1 of the Appendix. Some previously published data have been incorporated into the new data (Table A2, Appendix), and the Chapleau and Highway 101 data of Halls and Palmer (1990) have been split into three groups (C1, east; C2, central; C3, west) corresponding to the fault-bound units that they recognized within the southern KSZ. Overall mean directions for each area are given in Table 2. Only those site means for which $N \geq 5$ (including at least one end point and $\alpha_{95} \leq 20°$) are used to calculate the overall means, except for the relatively inferior MK data where $N \geq 2$. Although the MK mean values are less reliable than the other overall means, the small values of α_{95} suggest that they are not poor estimates of the true means.

Regional variations in paleomagnetism

Although the characteristic Matachewan paleomagnetic signature is clearly recovered from dykes in all parts of the swarm, the increased coverage afforded by the new data has led to the recognition of important variations in the direction of the remanence (Fig. 5). Statistical comparison of site mean data of opposite polarity in the same area, and of the same polarity from different areas (Table A3, Appendix) reveals significant differences at the 95% confidence level in both declination and inclination of the Matachewan remanence.

In all regions where N and R dykes occur together the N declination is displaced toward lower values compared with the antipodal R value. The displacement is a minimum (<5°) in the HP, HT, MK, and RA (Ranger Lake) areas and a maximum (5° → 10°) in the RL (Rocky Island Lake) and TM data sets (Tables 1 and 2). The N inclination is the same or greater than the corresponding R value, with a maximum departure (5° → 15°) in the TM, HT, MK, RL, and RA data sets. These data suggest reversal asymmetry similar to that described elsewhere from the Precambrian (Halls and Pesonen 1982), although for the HP data the N and R directions are almost exactly antipolar. While reversal asymmetry therefore cannot be ruled out, some other mechanism must have operated to produce the range of declinations and inclinations observed.

For any one polarity, the regional variations in inclination and declination do not show any obvious correlation but instead relate spatially, but in different ways, with the dyke swarm. We have therefore treated their analysis separately.

Variation in paleomagnetic inclination

A first-order observation of the regional data is that south of the KSZ R inclinations average −12° (range −6 to −16°) and about −23° (range −18 to −28°) within and to the north of the KSZ. The corresponding means for the N polarity are much closer at +24° and +28°. These differences are unlikely to be caused by deformation because the N polarity data have a virtually uniform inclination, and secondly because the R data would require differential crustal tilting of at least 10° over regions having dimensions of several hundred kilometres, leading to exposure of almost the full crustal thickness. The general sense of the required tilt, for example toward the north for data within and north of the KSZ, would indeed lead to higher grade rocks being exposed in the south, consistent with observation, but the near-vertical dip of the dykes and the relatively small gradient in regional metamorphism suggests that crustal tilting is at best only part of the explanation. A second interpretation is that the inclination change is a product of secular variation. We cannot discount this explanation except to note that for R dykes the extremes in inclination are well documented by many sites (high inclination areas OL, HP, HT, MK, and C (Chapleau) comprising 63 sites versus low-inclination areas RA, RL, TM, and MT with 80 sites) and that within each individual area the between-site variation (10–20°) is typical of that generally ascribed to secular variation in paleomagnetic studies.

A preferred solution is that the variation in inclination is the product of apparent polar wander during the evolution of the swarm. Either the magnetic pole moved relative to the Earth as a whole, possibly related to the polarity reversal, or the increasing inclination records a tectonic drift of Laurasia to higher latitudes. The range of observed inclination values (6°→32°) corresponds to a change in paleolatitude of 3–17°. The giant Mackenzie dyke swarm of northern Canada has areal dimensions

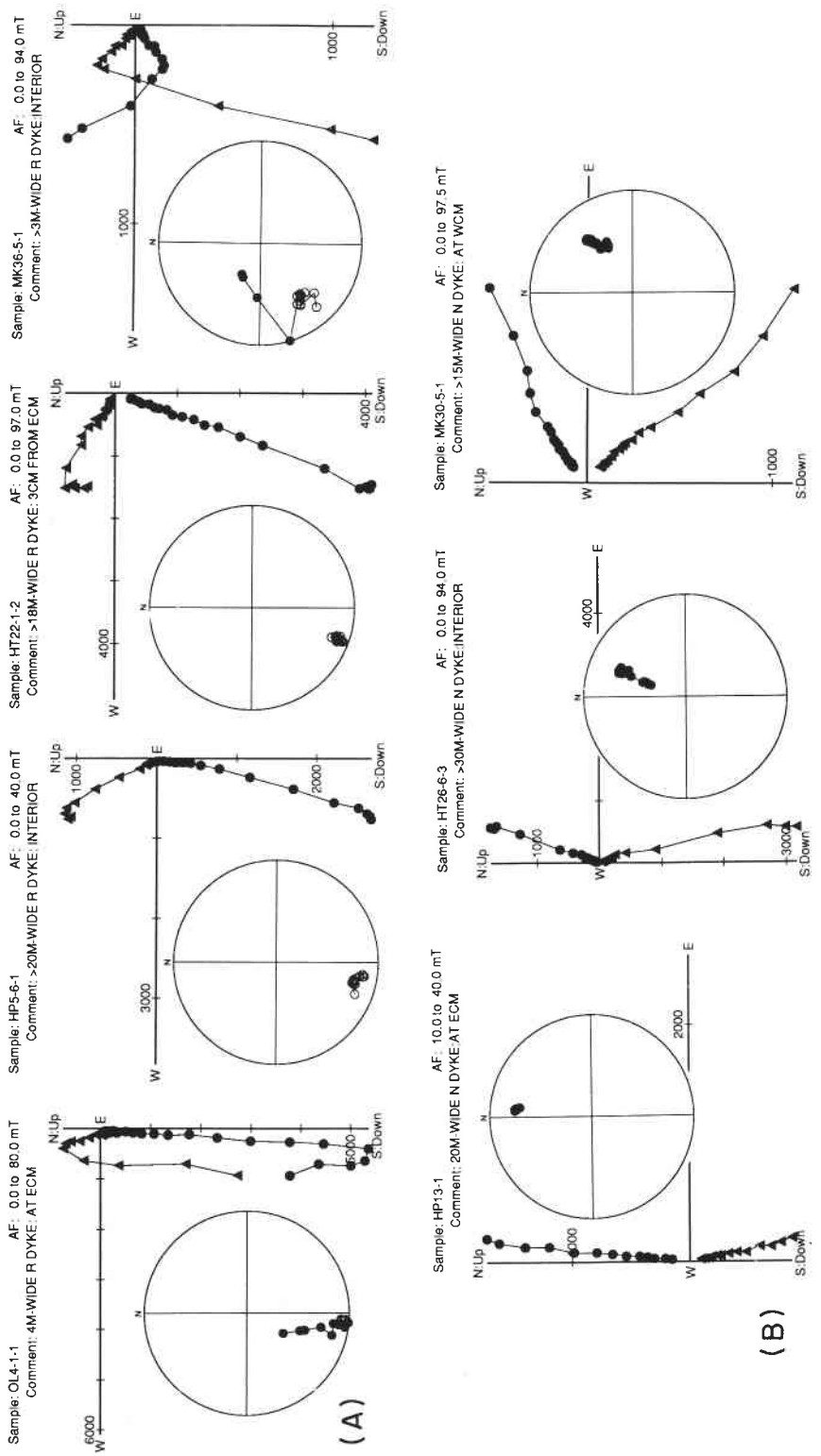


FIG. 4. Orthogonal vector plots and inset equal-area projections for Matachewan dyke samples after AF demagnetization. (A) R dykes from northwest of the KSZ. (B) N dykes from northwest of the KSZ. For both R and N dykes, note the increasing declination from left to right on approaching the KSZ from the northwest. (C) Data from the Timmins area, southwest of the KSZ, which serve as unrotated reference directions for R and N dykes. In the vector plots, triangles (circles) are projections of data onto the east–west vertical (horizontal) plane. In stereoplots, open (closed) symbols are upward (downward)-pointing magnetizations.

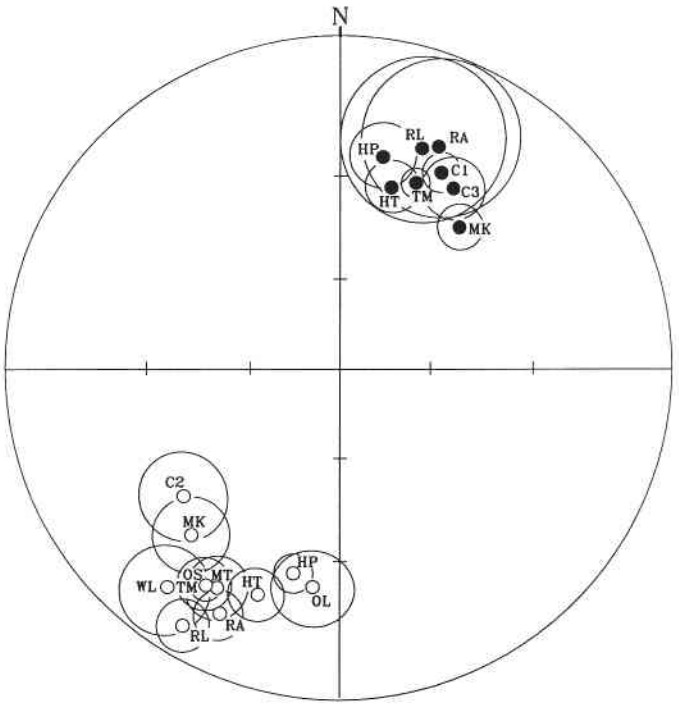


FIG. 5. Equal-angle stereographic projection of mean directions with circles of 95% confidence from this and previous studies. ●, lower hemisphere (N data); ○, upper hemisphere (R data). See Tables 1 and 2 for individual data points.

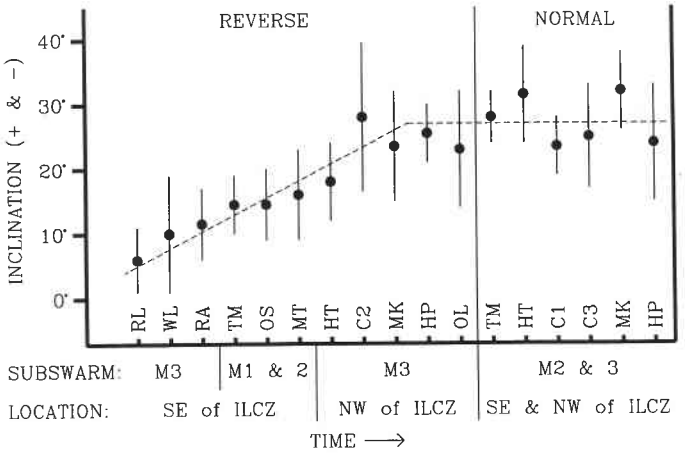


FIG. 6. Schematic diagram showing the observed change of magnetization inclination for the Matachewan swarm in space and possibly in time. It shows that towards the northwest the dykes acquired younger magnetizations either as a consequence of younger emplacement ages or because of uplift following delayed cooling at greater crustal depths. The model assumes a progressive change in inclination during the R period and a single reversal of the Earth's magnetic field during igneous activity (Halls 1991). Within the four subswarm groups no specific time sequence of the paleomagnetic data is implied. See Tables 1 and 2 for individual data points. Only those points are included for which $N \geq 3$ and $\alpha_{95} \leq 15°$. Error bars are taken from the α_{95} values. NOTE: C1 has been corrected for a 10° westward tilt of the dykes.

an order of magnitude greater than the Matachewan swarm and was emplaced over a period of only a few million years (Le-Cheminant and Heaman 1989). If the Matachewan swarm was emplaced within a similar time frame (Heaman 1989), the minimum drift rate of Laurasia would be 3°/Ma, which is comparable to previous estimates of 4–5°/Ma for particular time intervals in the Proterozoic (Irving 1979). From about 2.2 to 1.9 Ga ago mafic igneous rocks, represented by the Nipissing intrusions, and several dyke swarms in Ontario all yield high-latitude poles (Buchan and Halls 1990), so the movement of the Matachewan swarm may represent part of a trend toward higher paleolatitudes during early Proterozoic time.

If only a single field reversal (R⟶N) occurred during Matachewan igneous activity (Halls 1991), the polarity information represents a relative chronometer and can be used to trace the development of the swarm (Halls and Bates 1990). The general northwest increase in the ratio of N dykes to R dykes within the M3 swarm, for example, could indicate a general northward expansion of the swarm from the focal region. If the geomagnetic field became steeper during this time, then younger more distal R dykes would have higher inclinations and subsequent N dykes would have comparable inclinations. Thus the RL R data close to the focus record the lowest mean inclination, having been the site of the earliest, and possibly the longest, period (suggested by the high dispersion of the data) of dyke emplacement. This sequence for the M3 subswarm is shown schematically in Fig. 6. Because the ratio of N dykes to R dykes appears to increase with the level of regional metamorphism in the host rocks (Halls and Palmer 1990; Halls and Bates 1990), an alternative explanation for the relations in Fig. 6 is that they are produced by dykes at deeper crustal levels acquiring their magnetization at later times, after which differential crustal uplift displays the magnetization sequence at the surface. In this case younger dykes were either emplaced laterally as blade-like intrusions at successively greater depths (Halls and Bates 1990) or dykes were emplaced in an R field at low paleolatitudes, but at deeper crustal levels where ambient host rock temperatures approached the magnetite Curie temperature they did not finally acquire their magnetization (either R or N) on cooling until the continent had moved to higher paleolatitudes.

Variation in paleomagnetic declination

The spread of paleomagnetic declinations is greater than the spread of inclinations, with a range of about 40°. The statistical difference between most data sets (Table A3, Appendix) arises mainly because of different declination. As discussed above, the change in paleomagnetic direction could result from a shift of the geomagnetic pole relative to Laurasia. In the case of a geomagnetic excursion, unless movement of the pole was directly towards the dyke swarm, an accompanying change in observed declination would be expected, whereas a tectonic drift scenario would require a rotation of Laurasia as it moves to higher latitudes. While neither possibility can be discounted, we note that from region to region the change in declination does not smoothly follow the change in inclination, and that the N and R data from any one region have relatively antiparallel declinations. An unlikely sequence of drifting and rotation of Laurasia relative to the geomagnetic pole would be required in order to achieve the observed pattern.

A more plausible alternative is suggested when we consider the unusual configuration of the dyke swarm. The most significant changes in declination occur within and to the west of the KSZ, where the M3 subswarm describes a broad Z-pattern. Close to the KSZ, the dykes trend to the north and have a northeasterly N and southwesterly R declination. Toward the northwest the dykes swing around to a west-northwest strike and

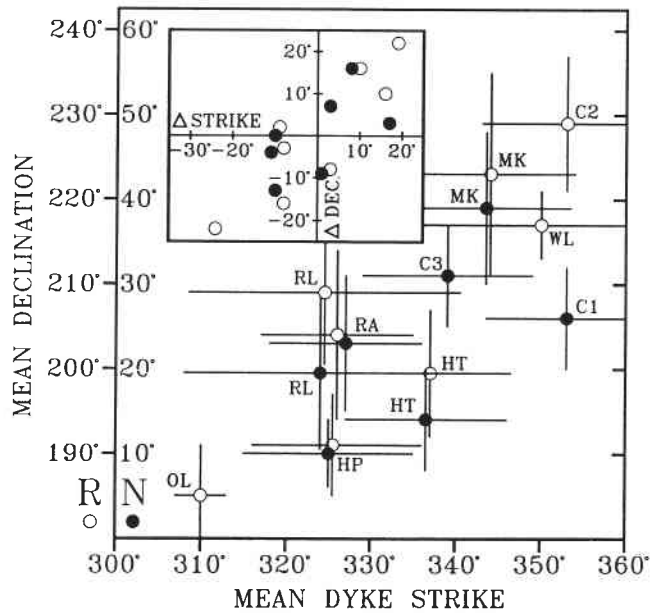


FIG. 7. Comparison of paleomagnetic declination and dyke strike for all data sets in the western or M3 subswarm. Bars are one standard deviation for both declination and dyke strike. N (●) and R (○) data are shown separately and slightly offset for clarity. The virtually constant N and R directions found across the Matachewan swarm within the Abitibi Subprovince suggest that the primary pattern of the dyke swarm was radial, not parallel, so although a positive correlation is apparent between declination and dyke strike for the M3 subswarm we would not expect a perfect correlation. Inset: data for the M3 subswarm plotted with respect to the mean values of declination and strike for the group of data sets; the N and R data are treated separately. The data are plotted relative to the mean (as Δ dec. and Δ strike) for comparison with Fig. 9 where individual areas show relatively little correlation.

TABLE 3. Estimated regional trends of the Matachewan swarm prior to distortion

Data set	Present strike (°)	Rotation (°) R data	Rotation (°) N data	Mean rotation (°)	Original trend (°)
OL	310	−23	—	−23	333
HP	326	−17	−11	−14	340
HT	337	−9	−7	−8	345
MK	344	+15	+18	+17	327
C3	339	—	+10	+10	329
C2	353	+21	—	+21	332
C1	353	—	+5	+5	348
WL	350	+8	—	+8	342
RL	325	+1	−2	0	325
RA	326	−4	+2	−1	327

NOTES: Correction of anomalous declinations west of and within the KSZ is with respect to the reference declination of 208° for R dykes and 21° for N dykes, based on TM, MT, and OS data from the M1 and M2 subswarms. Clockwise (anticlockwise) rotations from primary to *in situ* strike are positive (negative). Data sets are arranged in order of geographic location from northwest to southeast. The C1 dykes dip on average about 80°E and therefore may have suffered a westward component of tilt. This could reduce the estimated original trend by about 5°.

the declination also rotates anticlockwise. This positive correlation of mean declination and mean dyke trend (Fig. 7) would appear to be caused by gentle bending of the whole swarm since the time of emplacement, a possibility that was first raised by Ernst and Halls (1984) with more limited data. The initial configuration of the swarm is not known from independent evidence, so a formal paleomagnetic fold test (McFadden and Jones 1981) cannot be carried out. However, southeast of the KSZ, in the Abitibi Subprovince, all data sets (MT, OS (Otto stock), RL, and RA) have virtually identical mean site directions despite variations in dyke strike from 360° in the east to 325° in the west. Clearly, the dyke swarm has not been significantly disturbed, suggesting that a regionally important deformation of the Superior Province has not occurred in the Abitibi Subprovince and that the primary radial shape of the swarm has been preserved. Using the MT, OS, and TM data sets, we take 208° and 21° as the respective R and N declination reference values for the Matachewan swarm to which all other values of like polarity will be compared. Using this approach, dykes within and close to the KSZ appear to have been rotated clockwise (MK, C2, C3) up to 21° and those farther from the KSZ (HT, HP, OL) have suffered increasing amounts of anticlockwise rotation toward the northwest to a maximum of 23° (Table 3). After correction for rotation using our paleomagnetic data, the resulting dyke pattern (Fig. 8) is almost perfectly radial, becoming parallel farther from the swarm focus in the manner observed for smaller swarms centred on a single focus (e.g., Spanish Peaks, Colorado; Smith 1987).

Tests of the paleomagnetic data

Before accepting the thesis that significant vertical axis rotations of the Matachewan dyke swarm have occurred, several tests are made to ensure that the regional variations in direction are not the product of other unrelated effects such as contamination of the data by secondary components, anisotropy of remanence, or a statistical artifact. We consider each of these effects in turn.

Reliability of demagnetization data

Primary remanence is established in most sites by well-determined stable end points, but in light of variations in magnetic inclination and in particular a polarity asymmetry that could indicate an unremoved component such as a viscous one due to the present Earth's field (PEF), least-square planes were fitted through both sample level and site level data to test for systematic streaking.

For each data set, the poles to the planes fitted to the sample data for each site lie scattered along a plane perpendicular to the overall site mean direction, indicating no consistent streaking direction. Those sites with the most streaked distribution (estimated from the eigenvalues of the fitted plane) do not have the most anomalous site means. Likewise, sites distributed along a vertical plane do not have anomalous inclinations, nor do those distributed along a horizontal plane have anomalous declinations, although there is a suggestion of the former effect in the TM R data set. The same results were achieved by comparing planes fitted to the site means for each area and polarity. These observations indicate a lack of systematic streaking in the sample data attributable to the incomplete removal of a PEF or other component.

Magnetic anisotropy

Other possible causes of paleomagnetic variation are the effects of magnetic anisotropy. The magnetization of the Matachewan dykes could be deflected either as a result of intrinsic anisotropy of magnetic susceptibility of the diabase (Stacey 1960), or from shape anisotropy related to the tabular nature of dyke bodies (Abrahamsen 1986). In the first instance, anisotropy may arise from flow patterns in the dyke such that the maximum susceptibility axis lies within the dyke plane parallel to paleoflow, and the minimum axis is perpendicular to the dyke. However, typical susceptibilities are not high (0.05 SI units), and

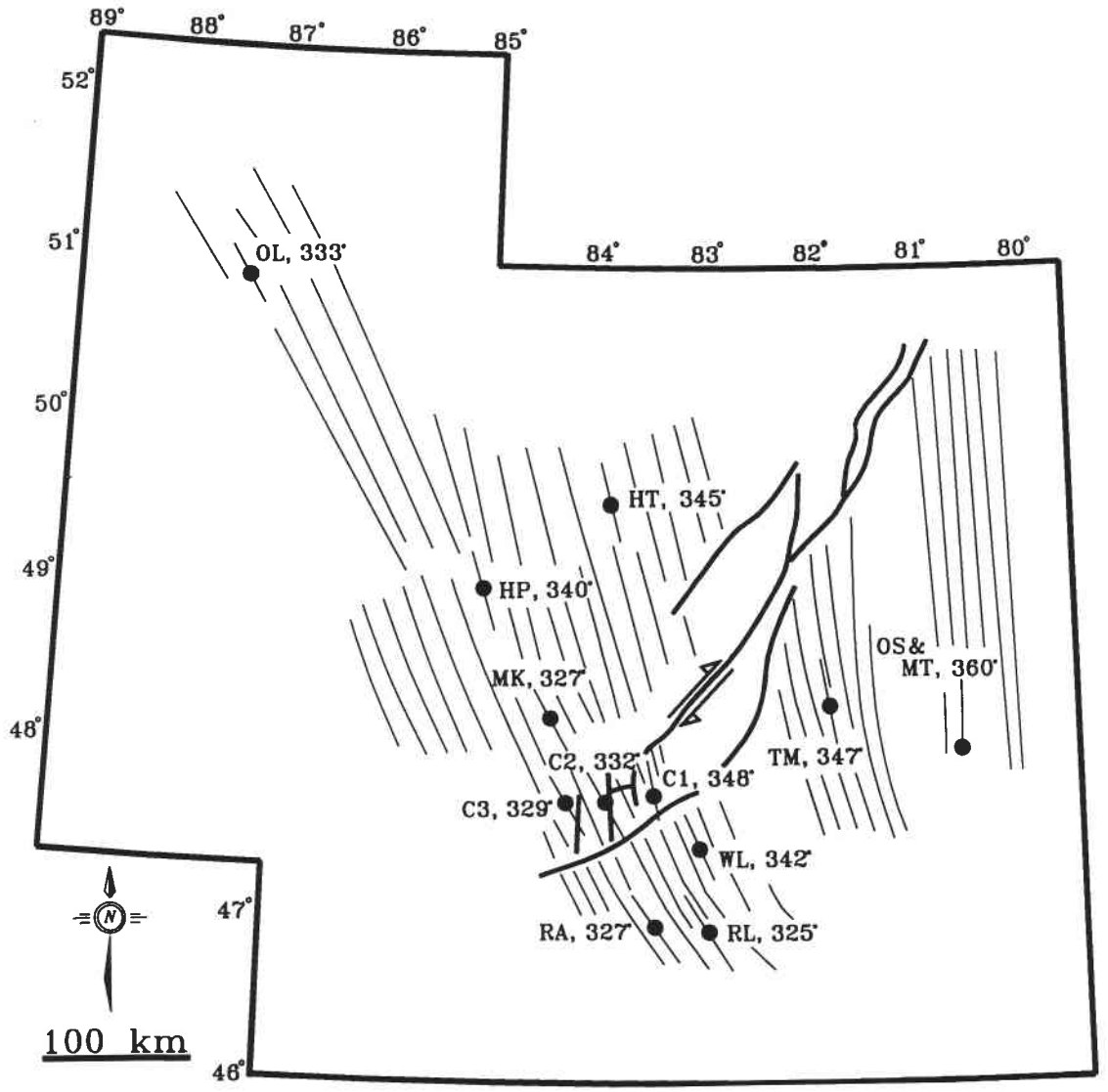


FIG. 8. The original pattern of the Matachewan dyke swarm generated using the procedure outlined in Table 3. The mean strike of the corrected dykes is given for each data set. The sense of strike-slip motion suggested by West and Ernst (1991), which may have offset the M3 subswarm by 60–80 km, is indicated.

low-field susceptibility anisotropy is too weak (<5%) to cause significant deflection of the ambient field (McElhinny 1973).

The effect of shape anisotropy, which is similar to refraction of light upon entering a different medium, can be estimated from the intensity of the natural remanent magnetization and the low-field susceptibility, which are used to calculate the apparent susceptibility (τ_{app}). For essentially vertical dykes, we would expect to see a deflection in the paleomagnetic declination towards the plane of the dyke, by an amount dependent on the angle between the paleofield and dyke plane. Values of τ_{app} ($\leq$0.25 SI units) calculated for the Matachewan dykes yield deflections based on Abrahamsen (1986) of less than 5° which are insufficient to explain the observed spread of declinations. However, due to varying degrees of low-grade alteration, we cannot be certain that the value of τ_{app} measured today is the same as that when the remanence was acquired. Therefore, a further empirical test was carried out by plotting paleomagnetic declination against dyke trend for each study area. The absence of any positive correlation (Fig. 9) shows that flow-induced or shape anisotropy is not significant. A similar plot using the different regional means (inset, Fig. 7) is clearly positive, suggesting that parts of the dyke swarm have rotated relative to one another.

Statistical comparison of data sets

The variations in Matachewan remanence appear to be real and significant. Statistical tests comparing the dispersion and mean directions (McFadden and Jones 1981; McFadden and Lowes 1981) for populations of sites from across the swarm have been carried out (Table A3, Appendix). First, the hypothesis that the populations share a common precision parameter (κ) is tested. The second test compares mean directions; where the populations have different values of κ, this test is only approximate. All data were recalculated for a common geographic location prior to comparison.

Grouping data into sets northwest of (HP, HT, OL), within (C1, C2, C3 $\pm$ MK), and southeast of (RA, RL, TM, WL (Wenebegon Lake)) the KSZ, we find significant differences at the 95% confidence level between the means for both polarities. However, statistically meaningful differences occur between the individual data sets too; the groups of sites come from separate subswarms, or different segments of the M3 subswarm along its

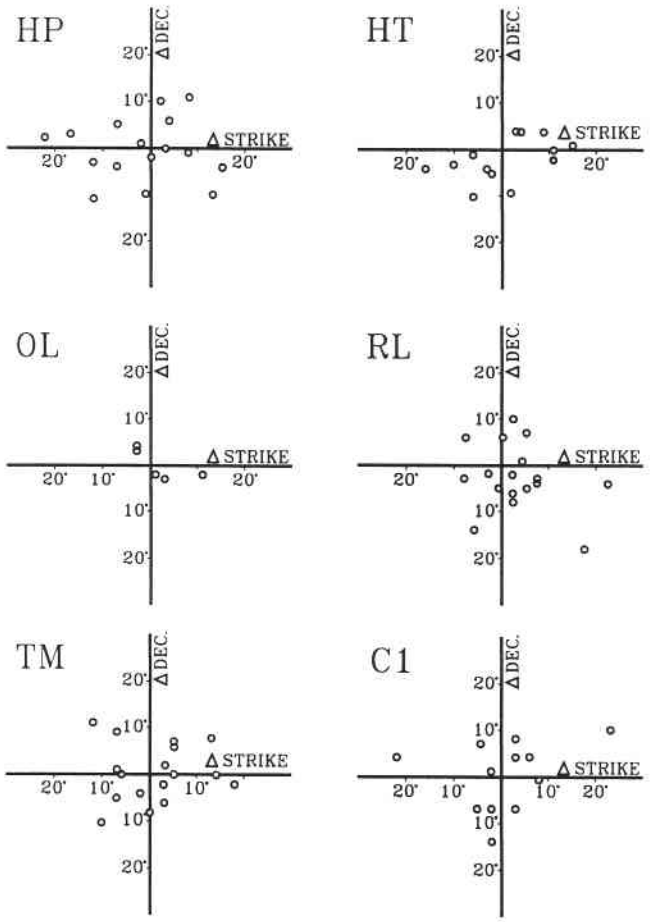


FIG. 9. Examples of variation in paleomagnetic declination (Δ dec.) against variation in dyke strike (Δ strike) for individual areas of the dyke swarm (see inset, Fig. 7 for explanation). No systematic relationship is apparent, indicating that fabric-induced or shape anisotropy of the dykes has not significantly deflected their magnetization.

strike. Data of opposite polarity from the same locality and of the same polarity but from different localities have been compared. In both instances statistically significant differences are found.

In some cases, we find that the populations do not share a common κ. For those tests involving the MK data, this is probably due to the lenient site selection criteria employed, resulting in a low precision. However, precision varies from low values (e.g., k (the estimate of κ) = 42 for RL R data) to high values (e.g., k = 430 for TM N data). The large number of sites in some of the low precision data sets suggests that low k values do not result from a failure to average secular variation. In fact, low k values may reflect longer periods of dyke emplacement and increased dispersion due to apparent polar wander.

In 80% of the tests, the hypothesis that the populations share a common mean could be rejected. This includes three out of the five tests between data of opposite polarity from the same locality, and reflects the asymmetry in the paleomagnetic direction. In some cases, the failure to distinguish separate populations probably results from the small number of sites in one or both of the data sets. In all other examples of a common mean, the data come from adjacent locations, where the remanence is the same or the difference is too small to detect. In summary, statistically significant variations in remanence are recorded between opposite polarities in single locations, and between different parts of the swarm for either polarity.

Discussion

Our paleomagnetic data suggest that the Matachewan swarm was injected during a period of continental movement to higher latitudes and was subsequently deformed northwest of and within the KSZ. If the swarm, as part of a moving plate, moved to higher latitudes away from the focal region and presumed site of the mantle plume source (Halls and Bates 1990) and if plumes remain fixed with respect to moving plates (Duncan and Richards 1991), later dykes might be expected to dominate *closer* to the focal region. If dyke remanences within and northwest of the KSZ are interpreted as being primary, in the sense that they date from the time of intrusion, then the northward increase in the proportion of N dykes would suggest the converse is true. However, the dilemma can be removed if most of the dykes were intruded at low paleolatitudes and the distribution of high-latitude remanences is the consequence of delayed remanence acquisition at deeper, hotter crustal levels during the growth of the swarm, followed by uplift. The prospect emerges, therefore, of dykes in which the remanence becomes progressively younger with increasing depth, changing from R polarity at the top to N polarity at the deepest crustal levels. The regions of greatest crustal uplift in the KSZ would then expose only N dykes, and regions of intermediate uplift to the north of the KSZ would show a more equal proportion of high-inclination N and R dykes, as observed. This interpretation allows the number of high-paleolatitude N and R dykes to be relatively small compared with that of low-latitude R dykes, a decline in dyke activity that would be expected because the minimum distance moved by the plate during dyke activity (about 1500 km) would presumably carry it off the plume.

The manner in which the Matachewan swarm was deformed may be similar to that suggested by paleomagnetic and structural studies in Phanerozoic terrains (Freund 1974; Luyendyk *et al.* 1980; Ron *et al.* 1984; McKenzie and Jackson 1986; Nur *et al.* 1989) where two-dimensional block rotations about more or less vertical axes arise from slip along sets of parallel fault planes that rotate in orientation, somewhat like a pack of cards. Similar block rotations are conceivable within the southwestern KSZ, accommodated by sinistral slip along a set of north–south faults that rotate clockwise, driven by northwest compression oblique to the faults. In the MK area north-northwest-trending dykes are locally sheared parallel to strike and show that tectonic activity postdated their emplacement.

Alternatively, rotation within the KSZ is related to strike slip parallel to the zone. Although sinistral motion along the KSZ (Watson 1980; Gibb 1983) has been proposed, dextral offset, which is supported by enhanced-resolution aeromagnetic data (West and Ernst 1991), is more in harmony with clockwise rotation within the zone.

Anticlockwise rotation of the dykes northwest of the KSZ may also be achieved by rotating fault orientations, such that dextral slip occurs along families of northeast-trending faults, but evidence for this faulting with the required frequency and offset to produce a 20° rotation is lacking. However, a similar mechanism involving relatively homogenous strain, as opposed to discrete fault slip, may be the answer. Some of the options are illustrated by West and Ernst (1991) (their Fig. 5); to be valid the mechanism must not produce any strong deflection of the east-northeast-trending Archean tectonic grain (Watson 1980) so clearly expressed by aeromagnetic maps of the central Superior Province. Furthermore, the aeromagnetic data show that the dykes are not buckled or boudinaged (Watson 1980), and apart from local shearing, particularly along margins, they lack a tectonic fabric; thus the dykes are probably close to their original

length. For these reasons we tentatively suggest that the anticlockwise rotation northwest of the KSZ was accommodated in the country rocks between the dykes by a combination of pure shear perpendicular to and simple shear parallel to (and possibly accommodated by) the preexisting Archean fabric in a transpressive regime. Maximum strains of <25% are required where the rotation is greatest (OL).

Our results indicate that early Proterozoic tectonism was not entirely restricted to the margins of the Superior Province and the KSZ. Independent evidence for tectonic activity in the region west of the KSZ is provided by reactivation of transcurrent faults at 1947 ± 23 Ma in the Rainy Lake region in the southwest Superior Province (Peterman and Day 1989).

The broad-scale deformation we record, including both differential uplift (Halls and Palmer 1990; Bates and Halls 1990) and vertical axis rotation, probably results from compression related to the Trans-Hudson orogeny during a major phase of KSZ uplift at 1.95 Ga. The notion that the KSZ might represent the foreland expression of the Trans-Hudson collision zone was first put forward by Burke (1983), and a model linking the KSZ via the Winisk River Fault to the formation of the Thompson Belt in Manitoba was proposed by Gibb (1983).

Although clear evidence of post-Matachewan tectonics exists within the KSZ, some uplift within this belt may predate the Matachewan dykes. For example, seismic reflection data suggest that the ILCZ and related faults are low-angle thrusts and that shortening across the KSZ exceeds 55 km (Geis *et al.* 1990). This deformation is not apparent southwest of the KSZ where the Matachewan swarm appears more or less continuous. Therefore the shortening appears to predate the intrusion of the swarm. However, the exact age, geological significance, or even reality of this earlier thrusting episode remains unclear.

Our paleomagnetic results have important consequences for the construction of APWP's. Formerly, cratonic areas were considered relatively immune from subsequent deformation, and so provided useful units within and between which paleomagnetic results could be compared. The Precambrian APWP for Laurentia has proved problematic to constrain, with only segments of the path truly well known (Buchan and Halls 1990). Much of the scatter of the data on the path may result from previously unrecognized structural rotations. Clearly, we have to be aware of tectonic processes after remanence acquisition that could significantly alter the position of a paleomagnetic pole, and in the light of our work some caution should be exercised in the evaluation of early Proterozoic data from Archean shields. Fortunately for the Superior Province, most of the data come from southeast of the KSZ where the Matachewan swarm has not been disturbed, so that errors introduced in the APWP by distortions of the type suffered by the swarm elsewhere are likely to be insignificant.

Conclusions

A characteristic high-coercivity, dual polarity, primary remanence has been recovered from the entire Matachewan dyke swarm. Although the signature is readily recognized as Matachewan, significant variations occur in both declination and inclination of the magnetization. The inclination changes are most easily explained in terms of either apparent or true wander of the geomagnetic pole, such that later dykes were intruded in a 20° steeper ambient paleofield. Declination changes appear to result from distortion of the dyke swarm across and west of the KSZ, resulting in approximately 20° rotations both clockwise and anticlockwise relative to the undeformed sector of the swarm in the Abitibi Subprovince. Our study shows therefore that Proterozoic deformation is not necessarily restricted to the margins of, or along discrete zones within, Archean cratons, but that the effects of marginal orogenies may penetrate into the interiors of ancient shields in a more pervasive manner than hitherto recognized.

Acknowledgments

The authors appreciate the comments of L. Pryer and G. F. West (University of Toronto) and in particular R. Ernst (Ottawa–Carleton Geoscience Centre). A. Nur (Stanford University) provided us with a preprint of his paper on block rotations. C. Dodds, R. McGrory, M. Lawton, and S. Pfleiderer (all University of Toronto) acted as field and laboratory assistants. D. G. Ball helped to draft several of the figures and also devised the computer plotting program on which Fig. 4 is based. A reviewer is thanked for raising issues concerning the growth of a dyke swarm as it moves away from its parent plume. This research was financed by grants from the Lithoprobe Supporting Geoscience Projects Program and by Natural Sciences and Engineering Research Council of Canada Operating Grant A7824.

ABRAHAMSEN, N. 1986. On shape anisotropy. *In* Twenty-five years of geology in Aarhus. *Edited by* J. T. Möller. Geoskrifter, no. 24, pp. 11–21.

AIBANGBEE, F. O. 1982. The paleomagnetism of some diabase dykes north of Elliot Lake, Ontario. M.Sc. thesis, University of Western Ontario, London, Ont.

BAILEY, R. C., and HALLS, H. C. 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics, **54**: 174–182.

BATES, M. P., and HALLS, H. C. 1990. Regional variation in paleomagnetic polarity of the Matachewan dyke swarm related to the Kapuskasing Structural Zone, Ontario. Canadian Journal of Earth Sciences, **27**: 200–211.

BUCHAN, K. L., and HALLS, H. C. 1990. Paleomagnetism of Proterozoic mafic dyke swarms of the Canadian Shield. *In* Mafic dykes and emplacement mechanisms. *Edited by* A. J. Parker, P. C. Rickwood and D. H. Tucker. Balkema, Rotterdam, pp. 209–230.

BUCHAN, K. L., NEILSON, D. J., and HALE, C. J. 1990. Relative age of Otto stock and Matachewan dykes from paleomagnetism and implications for the Precambrian polar wander path. Canadian Journal of Earth Sciences, **27**: 915–922.

BURKE, K. 1983. The Kapuskasing Structure. *In* Workshop on a cross section of Archean crust. *Edited by* L. D. Ashwal and K. D. Card. Lunar and Planetary Institute, Technical Report 83-03, pp. 21–23.

CARD, K. D., PERCIVAL, J. A., LAFLEUR, J., and HOGARTH, D. D. 1981. Progress report on regional geological synthesis, central Superior Province. *In* Current research, part A. Geological Survey of Canada, Paper 81-1A, pp. 77–93.

CONDIE, K. C., BOBROW, D. J., and CARD, K. D. 1987. Geochemistry of Precambrian mafic dykes from the southern Superior Province of the Canadian Shield. *In* Mafic dyke swarms. *Edited by* H. C. Halls and W. F. Fahrig. Geological Association of Canada, Special Paper 34, pp. 95–108.

COOK, F. A. 1985. Geometry of the Kapuskasing Structure from a Lithoprobe pilot reflection survey. Geology, **13**: 368–371.

DUNCAN, R. A., and RICHARDS, M. A. 1991. Hotspots, mantle plumes, flood basalts, and true polar wander. Reviews of Geophysics, **29**: 31–50.

ERNST, R. E., and HALLS, H. C. 1984. Paleomagnetism of the Hearst dike swarm and implications for the tectonic history of the Kapuskasing Structural Zone, northern Ontario. Canadian Journal of Earth Sciences, **21**: 1499–1506.

FAHRIG, W. F. 1987. The tectonic settings of continental mafic dyke swarms: failed arm and early passive margin. *In* Mafic dyke swarms. *Edited by* H. C. Halls and W. F. Fahrig. Geological Association of Canada, Special Paper 34, pp. 331–348.

FAHRIG, W. F., GAUCHER, E. H., and LAROCHELLE, A. 1965. Palaeomagnetism of diabase dykes of the Canadian shield. Canadian Journal of Earth Sciences, **2**: 278–298.

FREUND, R. 1974. Kinematics of transform and transcurrent faults. Tectonophysics, **21**: 93–134.

GEIS, W. T., COOK, F. A., GREEN, A. G., MILKEREIT, B., PERCIVAL, J. A., and WEST, G. F. 1990. Thin thrust sheet formation of the Kapuskasing Structural Zone revealed by Lithoprobe seismic reflection data. Geology, **18**: 513–516.

GIBB, R. A. 1983. Model for the suturing of Superior and Churchill plates: An example of double indentation tectonics. Geology, **11**: 413–417.

HALLS, H. C. 1986. Paleomagnetism, structure, and longitudinal correlation of Middle Precambrian dykes from northwestern Ontario and Minnesota. Canadian Journal of Earth Sciences, **23**: 142–157.

——— 1991. The Matachewan dyke swarm, Canada: an early Proterozoic magnetic field reversal. Earth and Planetary Science Letters, **105**: 279–292.

HALLS, H. C., and BATES, M. P. 1990. The evolution of the 2.45 Ga Matachewan dyke swarm, Canada. *In* Mafic dykes and emplacement mechanisms. *Edited by* A. J. Parker, P. C. Rickwood and D. H. Tucker. Balkema, Rotterdam, pp. 237–250.

HALLS, H. C., and PALMER, H. C. 1990. The tectonic relationship of two Early Proterozoic dyke swarms to the Kapuskasing Structural Zone: a paleomagnetic and petrographic study. Canadian Journal of Earth Sciences, **27**: 87–103.

HALLS, H. C., and PESONEN, L. J. 1982. Paleomagnetism of Keweenawan Rocks. *In* Geology and tectonics of the Lake Superior region. *Edited by* R. J. Wold and W. J. Hinze. Geological Society of America, Memoir 156, pp. 173–201.

HALLS, H. C., and SHAW, E. G. 1988. Paleomagnetism and orientation of Precambrian dykes, eastern Lake Superior region, and their use in estimates of crustal tilting. Canadian Journal of Earth Sciences, **25**: 732–743.

HALLS, H. C., SHAW, E. G., ERNST, R. E., BISSON, J. R., and RANALLI, G. 1985. Regional structural changes of the Matachewan–Hearst dyke swarm, northern Ontario, and their tectonic implications [abstract]. International Conference on Mafic Dyke Swarms, Toronto, Program with Abstracts, pp. 75–78.

HANES, J. A., ARCHIBALD, D. A., QUEEN, M., and LEE, J. K. W. 1988. Ar40/Ar39 geochronology of diabase dykes: implications for the tectonothermal evolution of the Kapuskasing uplift. Lithoprobe workshop on the Kapuskasing Structural Zone I, Toronto, pp. 7–16.

HEAMAN, L. M. 1989. U–Pb dating of mafic dyke swarms: what are the options? *In* International Association of Volcanology and Chemistry of the Earth's Interior General Assembly, Program of Abstracts. New Mexico Bureau of Mines and Mineral Resources, Bulletin 131, p. 125.

HOFFMAN, P. F. 1988. United plates of America, the birth of a craton. Annual Review of Earth and Planetary Sciences, **16**: 543–603.

IRVING, E. 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Canadian Journal of Earth Sciences, **16**: 669–694.

IRVING, E., and NALDRETT, A. J. 1977. Paleomagnetism in Abitibi greenstone belt and Abitibi and Matachewan diabase dikes: Evidence of the Archean geomagnetic field. Journal of Geology, **85**: 157–176.

KIRSCHVINK, J. L. 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

KISSEL, C., and LAJ, C. 1989. Paleomagnetic rotations and continental tectonics. Kluwer, Dordrecht.

LECHEMINANT, A. N., and HEAMAN, L. M. 1989. Mackenzie igneous events, Canada: Middle Proterozoic hotspot magmatism associated with ocean opening. Earth and Planetary Science Letters, **96**: 38–48.

LUYENDYK, B. P., KAMERLING, M. J. and TERRES, R. 1980. Geometric model for Neogene crustal rotations in southern California. Geological Society of America Bulletin, **91**: 211–217.

MCELHINNY, M. W. 1973. Palaeomagnetism and plate tectonics. Cambridge University Press, Melbourne.

MCFADDEN, P. L., and JONES, D. L. 1981. The fold test in palaeomagnetism. Geophysical Journal of the Royal Astronomical Society, **67**: 53–58.

MCFADDEN, P. L., and LOWES, F. J. 1981. The discrimination of mean directions drawn from Fisher distributions. Geophysical Journal of the Royal Astronomical Society, **67**: 19–33.

MCKENZIE, D., and JACKSON, J. 1986. A block model of distributed deformation by faulting. Journal of the Geological Society (London), **143**: 349–353.

NUR, A., RON, H., and SCOTTI, O. 1989. Kinematics and mechanics of tectonic block rotations. *In* Deformation and transmission of stress in the Earth. *Edited by* S. C. Cohen and P. Vanicek. American Geophysical Union, Monograph 49, pp. 31–46.

PALMER, H. C., and BARNETT, R. L. 1988. Amphibole chemistry of Matachewan and Kapuskasing dikes and its bearing on the timing of Kapuskasing Structural Zone uplift. Lithoprobe workshop on the Kapuskasing Structural Zone II, Toronto, pp. 185–187.

PERCIVAL, J. A. 1983. High-grade metamorphism in the Chapleau-Foleyet area, Ontario. American Mineralogist, **68**: 667–686.

PERCIVAL, J. A., and CARD, K. D. 1985. Structure and evolution of Archean crust in central Superior Province, Canada. *In* Evolution of Archean supracrustal sequences. *Edited by* L. D. Ayers, P. C. Thurston, K. D. Card and W. Weber. Geological Association of Canada, Special Paper 28, pp. 179–192.

PERCIVAL, J. A., PARRISH, R. R., KROGH, T. E., and PETERMAN, Z. E. 1988. When did the Kapuskasing Zone come up? Lithoprobe workshop on the Kapuskasing Structural Zone I, Toronto, pp. 43–47.

PETERMAN, Z. E., and DAY, W. 1989. Early Proterozoic activity on Archean faults in the western Superior Province—Evidence from pseudotachylyte. Geology, **17**: 1089–1092.

PHINNEY, W. C., MORRISON, D. A., and MACZUGA, D. E. 1988. Anorthosites and related megacrystic units in the evolution of Archean crust. Journal of Petrology, **29**: 1283–1323.

PULLAIAH, G., IRVING, E., BUCHAN, K. L., and DUNLOP, D. J. 1975. Magnetization changes caused by burial and uplift. Earth and Planetary Science Letters, **28**: 133–143.

RON, H., FREUND, R., GARFUNKEL, Z., and NUR, A. 1984. Block rotation by strike slip faulting: structural and paleomagnetic evidence. Journal of Geophysical Research, **89**: 6256–6270.

SCHUTTS, L. D., and DUNLOP, D. J. 1981. Proterozoic magnetic overprinting of Archaean rocks in the Canadian Superior Province. Nature (London), **291**: 642–645.

SMITH, R. P. 1987. Dyke emplacement at Spanish Peaks, Colorado. *In* Mafic dyke swarms. *Edited by* H. C. Halls and W. F. Fahrig. Geological Association of Canada, Special Paper 34, pp. 47–54.

STACEY, F. D. 1960. Magnetic anisotropy of igneous rocks. Journal of Geophysical Research, **65**: 2429–2442.

STRANGWAY, D. W. 1964. Rock magnetism and dyke classification. Journal of Geology, **72**: 648–663.

THURSTON, P. C., SIRAGUSA, G. M., and SAGE, R. P. 1977. Geology of the Chapleau area, Districts of Algoma, Sudbury and Cochrane. Ontario Division of Mines, Geology Report 157.

VAN DER VOO, R., and CHANNEL, J. E. T. 1989. Paleomagnetism in orogenic belts. Reviews of Geophysics and Space Physics, **18**: 455–481.

WATSON, J. 1980. The origin and history of the Kapuskasing structural zone, Ontario, Canada. Canadian Journal of Earth Sciences, **17**: 866–875.

WEST, G. F., and ERNST, R. E. 1991. Evidence from aeromagnetics on the configuration of Matachewan dykes and the tectonic evolution of the Kapuskasing Structural Zone, Ontario, Canada. Canadian Journal of Earth Sciences, **28**: 1797–1811.

ZIJDERVELD, J. D. A. 1967. A.C. demagnetization of rocks: analysis of results. *In* Methods in palaeomagnetism. *Edited by* D. W. Collinson, K. M. Creer, and S. K. Runcorn. Elsevier, Amsterdam, pp. 254–286.

Appendix

TABLE A1. Summary of new paleomagnetic data from the Matachewan dyke swarm

Site[a]	Width (m)	Dyke strike (°)	Dyke dip (°)	EP[b]	RC[c]	RS[d]	Declination (°)	Inclination (°)	k[e]	α_{95}[f] (°)	P[g]
HP3[h]	36.0	320	90	3	7	2	196.7	−18.7	84	7.2	R
HP5[h]	>20.0	320	—	6	—	—	190.3	−23.6	103	6.6	R
HP6[h]	>19.9	325	85W	2	3	2	192.9	−32.6	140	9.5	R
HP7[h]	10.2	327	90	8	—	—	189.9	−30.9	260	3.5	R
HP8[h]	2.7	335	85W	6	1	—	10.9	22.4	170	4.9	N
HP9[h]	10.0	331	90	5	2	—	198.3	−28.2	270	4.1	R
HP10[h]	10.0	315	85W	8	—	—	188.7	−25.0	313	3.1	R
HP11[h]	13.4	340	83E	7	—	—	182.2	−23.3	480	2.8	R
HP12[h]	1.0	326	80W	7	—	—	181.5	−33.9	210	4.2	R
HP13[h]	20.3	342	85W	7	—	—	7.6	29.6	350	3.2	N
HP14[h]	21.5	306	85W	6	2	—	193.6	−26.3	230	4.0	R
HP15[h]	15.8	329	90	1	4	4	202.0	−19.6	93	15.0	R
HP16	27.0	307	86E	—	—	6	—	—	—	—	—
HP17	28.0	308	80E	3	—	—	191.6	−33.0	1900	2.8	R
HT1[h]	7.6	340	84E	5	—	—	202.4	−22.3	170	5.8	R
HT2[h]	28.0	323	—	3	3	—	213.6	−6.1	43	16.0	R
HT3	>17.0	318	84W	2	3	1	11.8	21.4	8	48.0	N
HT3	>17.0	318	84W	3	—	3	202.6	−31.6	31	22.5	R
HT4[h]	7.5	334	90	5	—	1	193.4	−26.5	82	8.5	R
HT6[h]	48.0	335	—	5	—	—	9.3	29.3	89	8.2	N
HT7	>41.0	310	—	4	—	—	14.8	22.7	410	4.5	N
HT8[h]	>19.0	—	—	5	—	—	5.8	41.5	120	7.1	N
HT9	>11.0	—	—	1	—	3	192.4	−19.3	—	—	R
HT10[h]	>48.0	328	—	5	—	—	10.3	31.1	600	3.1	N
HT11[h]	>18.0	347	—	4	2	—	198.4	9.8	48	12.0	R
HT12[h]	55.0	347	63W	5	—	—	16.9	23.9	180	5.8	N
HT13[h]	39.0	330	90	5	—	1	197.0	−10.3	490	3.5	R
HT14[h]	40.0	340	90	5	—	—	15.9	28.5	86	8.3	N
HT15	>47.0	340	—	—	2	3	209.7	−39.9	—	—	R
HT16	>5.0	—	—	—	3	—	203.6	71.2	280	—	R
HT17	>29.0	345	85W	1	2	3	190.6	−17.9	26	68.0	R
HT18[h]	>14.0	320	—	5	—	—	193.6	−24.3	110	7.3	R
HT19[h]	>40.0	330	—	5	—	—	187.5	−26.9	72	9.1	R
HT20[h]	40.0	339	90	5	—	—	201.9	−20.8	97	7.8	R
HT21[i]	>2.2	342	—	3	—	—	301.8	57.8	210	8.4	
HT22[h]	>18.0	347	85E	6	—	—	196.0	−13.7	280	4.0	R
HT23[h]	>13.5	351	—	5	—	—	199.4	−16.5	99	7.7	R
HT24[i]	0.5	348	90	5	—	—	116.7	−43.2	53	10.6	
HT25[i]	>2.0	005	—	1	—	4	1.5	41.5	—	—	
HT26[h]	>25.0	—	—	3	2	—	26.6	34.4	81	12.0	N
HT27	>19.5	—	—	2	3	1	205.9	−28.4	6	56.0	R
HT28	>4.5	351	90	2	1	—	190.0	−22.8	79	28.4	R
MK1	28.0	330	—	1	—	3	197.5	−34.6	—	—	R
MK2E[i]	8.0	335	90	—	—	2	—	—	—	—	
MK2W[i]	>13.5	358	78W	—	—	3	—	—	—	—	
MK3[i]	24.0	340	—	—	—	2	—	—	—	—	
MK4	17.0	325	85W	1	—	5	31.1	11.7	—	—	N
MK5[j]	7.0	355	80W	2	—	3	46.6	40.6	40	40.7	N
MK6[i]	55.0	355	—	—	—	4	—	—	—	—	
MK8E[i]	20.0	348	80W	—	—	2	—	—	—	—	
MK8W[j]	>6.0	355	—	4	—	2	192.4	−10.2	14	25.3	R
MK9[i]	18.5	340	85W	—	—	4	—	—	—	—	
MK10[i]	>8.0, <20.0	320	70W	—	—	3	—	—	—	—	
MK11[j]	23.0	335	70W	3	—	1	40.6	40.7	14	33.9	N
MK12E[j]	18.0	350	85W	4	—	2	18.6	36.1	17	23.1	N
MK13	13.5	338	85W	—	—	6	—	—	—	—	R?
MK14	20.0	355	80W	2	—	3	35.8	47.8	20	58.6	N
MK15	10.0	343	80W	1	—	4	31.4	20.5	—	—	N
MK16E[i]	15.0	314	80W	—	—	4	—	—	—	—	
MK16W	4.5	320	90	1	—	2	208.9	−29.7	—	—	R

TABLE A1 (*continued*)

Site[a]	Width (m)	Dyke strike (°)	Dyke dip (°)	EP[b]	RC[c]	RS[d]	Declination (°)	Inclination (°)	k[e]	α_{95}[f] (°)	P[g]
MK17[j]	1.3	335	84E	5	—	—	35.2	22.1	95	7.1	N
MK18	>11.5,<18.5	335	87W	1	—	2	253.0	−29.6	—	—	R?
MK19[j]	19.0	347	88E	3	—	3	212.7	−27.3	330	6.8	R
MK20	25.0	343	90	—	—	3	—	—	—	—	R?
MK21	18.6	344	82E	4	—	1	221.2	−55.6	37	15.0	R?
MK22[i]	4.7	311	90	—	—	2	—	—	—	—	
MK23[j]	>18.0	340	—	2	2	2	235.9	−38.2	36	24.0	R
MK24[i]	>1.0	346	90	—	—	3	—	—	—	—	
MK25[i]	>10.0	353	88E	—	—	3	—	—	—	—	
MK26	>6.2	319	85E	—	—	3	—	—	—	—	R?
MK27E	1.3	344	85E	1	—	2	199.7	−21.4	—	—	R
MK27W[j]	1.1	327	83E	5	—	—	34.2	50.0	74	8.9	N
MK28[j]	>21.5	343	83E	4	—	—	48.4	35.8	83	10.2	N
MK29	>29.0	328	90	—	—	6	—	—	—	—	N?
MK30[j]	—	331	87W	3	—	2	54.8	30.5	96	12.6	N
MK31[j]	>14.9	326	81E	2	—	4	221.5	−33.6	48	36.8	R
MK32[i]	21.5	342	90	—	—	4	—	—	—	—	
MK33[j]	>14.0	342	90	2	1	2	40.5	28.3	140	15.0	N
MK34[j]	>5.2	334	87E	3	—	—	41.9	26.1	28	23.8	N
MK35	>12.0, <24.0	339	86E	1	—	2	29.8	34.9	—	—	N
MK36[j]	>3.0	—	—	6	—	—	231.6	−29.6	61	8.7	R
MK37[j]	—	340	—	4	—	1	223.1	−19.7	28	17.6	R
MK38[j]	>10.0	354	—	3	—	2	24.2	27.5	60	16.0	N
MK39[j]	48.0	337	—	2	—	2	246.1	−20.0	66	31.1	R
MK40[j]	22.0	354	86E	2	—	4	226.5	−30.8	71	30.1	R
MK41	>20.0	343	90	1	—	4	44.3	20.9	—	—	N?
MK42[i]	15.5	004	90	—	—	3	—	—	—	—	
MK43	25.0	348	68E	1	—	4	223.7	−26.1	—	—	R
MK44	17.5	13	78E	1	—	4	220.6	−15.1	—	—	R
MK45	19.0	354	85W	—	—	5	—	—	—	—	R?
MK46E	1.5	342	83W	1	1	4	251.6	−33.9	4800	16.0	R
MK46W	>16.5	341	81E								
MK47[i]	9.0	345	75W	—	—	3	—	—	—	—	
OL1[h]	34.0	307	90	7	—	3	188.2	−20.4	140	5.1	R
OL2[h]	33.5	311	90	6	—	3	182.7	−29.2	38	11.0	R
OL3[h]	21.0	307	90	5	—	2	189.3	−28.2	24	15.9	R
OL4[h]	4.2	313	77W	5	1	—	181.7	−14.3	110	7.1	R
OL5	>7.0	302	80W	3	—	4	196.6	−16.3	830	4.3	R
TM1[h]	>34.0	007	90	5	—	—	16.6	27.9	210	5.3	N
TM2[h]	21.5	010	90	5	—	2	21.0	27.0	82	8.5	N
TM3[h]	>20.0	000	—	5	—	—	207.3	−16.9	61	9.9	R
TM4	5.0	330	87W	4	—	1	25.5	39.6	68	11.2	N
TM5	>20.0	336	90	3	—	2	201.3	−16.0	50	17.7	R
TM6	5.2	325	89W	2	—	1	28.4	25.6	3300	4.4	N
TM7	>1.0	320	85W	1	—	3	38.7	14.2	—	—	N
TM8[h]	20.5	355	90	5	—	1	216.7	−21.7	37	12.8	R
TM9	25.0	351	85W	4	—	2	215.2	−12.0	37	15.2	R
TM10	>3.0	359	90	1	—	—	211.2	18.1	—	—	R
TM11[i]	15.0	000	—	—	3	—	252.3	79.6	34	9.9	
TM12[h]	>7.0	345	82E	4	1	—	211.4	−16.1	150	7.3	R
TM13[h]	>8.0	341	82E	5	—	—	216.5	−11.1	69	9.3	R
TM14[h]	>11.0	340	85E	4	1	—	204.6	−27.6	38	14.0	R
TM15[h]	6.5	345	89W	4	1	—	203.4	−12.1	43	14.0	R
TM16[h]	29.0	336	83W	5	—	1	209.1	−7.4	52	11.7	R
TM17	18.0	349	83W	8	—	4	206.6	−3.6	64	7.0	R
TM18[h]	13.0	339	86E								
TM19	6.0	347	90								
TM20	34.0	018	79E	3	—	1	212.2	−15.1	920	4.1	R
TM21[h]	42.0	001	90	5	—	1	27.6	29.5	120	7.0	N
TM22[h]	19.0	335	78E	5	—	—	217.6	−11.7	37	12.8	R
TM23	4.0	355	85W	—	—	—	—	—	—	—	—
TM24	18.0	334	90	2	—	1	25.8	38.1	75	29.2	R

TABLE A1 (*concluded*)

Site[a]	Width (m)	Dyke strike (°)	Dyke dip (°)	EP[b]	RC[c]	RS[d]	Declination (°)	Inclination (°)	k[e]	α_{95}[f] (°)	P[g]
TM25	10.0	000	—	3	—	—	222.5	−13.6	71	14.7	R
TM26	5.0	000	—	—	—	—	—	—	—	—	—
TM27[h]	0.3	332	80W	5	—	—	198.6	−16.6	120	7.1	R
TM28[h]	12.0	347	90	5	—	—	214.5	−14.1	47	11.2	R
TM29E/29W	15.4	319	90	3	—	9	23.4	35.3	53	17.2	N
TM29E/29W[h]	15.5	004	60E	4	1	6	200.9	−13.0	110	8.6	R
TM30[h]	>8.5	333	90	3	—	—	198.9	−16.1	47	18.1	R
TM31[h]	17.0	335	88E	3	2	4	19.3	27.0	280	6.3	N
TM31[h]	17.0	335	88E	9	—	—	204.2	−19.7	39	8.3	R
TM32	8.5	338	68E	1	2	9	23.0	31.0	5300	4.6	N
TM33[h]	3.2	349	86E	7	—	5	208.9	−19.5	130	5.4	R
TM34	16.0	355	79E								
TM35	>2.5, <15.0	356	84E	4	—	—	205.5	−12.1	140	7.9	R
TM36	2.3	352	87E	—	—	—	—	—	—	—	—
TM37	>11.0, <26.0	339	89E	4	—	—	221.0	−13.6	57	12.2	R
TM38	—	337	90	—	—	—	—	—	—	—	—
TM39[h]	17.0	330	85W	5	—	—	219.9	−10.3	62	9.8	R
TM40	4.8	355	80E	3	—	—	212.3	−0.4	27	24.2	R
TM41	10.5	337	78E	4	—	1	204.9	−11.6	75	10.7	R
TM42[h]	>12.0, <15.0	349	90	5	—	—	22.1	29.5	140	6.6	N
TM43	26.5	341	—	3	—	3	20.5	23.6	21	27.6	N
TM43	26.5	341	—	3	—	3	207.7	−10.0	34	21.5	R
TM44	1.7	000	85W	3	2	—	218.0	−7.7	20	23.0	R

[a]HP, Hornepayne; HT, Hearst; MK, Whitefish, Manitowik, Matchinameigus, and Dog lakes; OL, Ogoki Lake; and TM, Timmins. Site HT3 is the same location as site E4 of Ernst and Halls (1984). Site HP17 is not shown in Fig. 3, but is located on Highway 17 about 2 km east of the junction with Highway 614. For many MK sites where acceptable stable end points were not achieved, the polarity of the dyke at least could be inferred.

[b]EP, number of samples that gave stable end points.

[c]RC, number of samples that gave remagnetization circles.

[d]RS, number of samples that failed to provide a Matachewan remanence vector.

[e]k, estimate of precision parameter.

[f]α_{95}, semiangle of the cone of 95% confidence.

[g]P, magnetic polarity: N, normal; R, reverse.

[h]Site used to calculate mean directions for the different data sets ($N \geq 5$ including at least one end point, and $\alpha_{95} \leq 20°$).

[i]Site that failed to yield a recognizable Matachewan remanence.

[j]Site used to calculate the mean direction for the MK data set ($N \geq 2$).

TABLE A2. The contribution of previously published data to the regional paleomagnetic mean directions in Table 2

Mean	New sites	Published data by site	Reference
HP	HP3, HP5–17	E11, 12	Ernst and Halls 1984
MK	MK1–47	KS2–11	Halls and Palmer
C1		KS1, E18–19, 159–161, 173 174, 217–227, 231–233, HP17	Halls and Palmer 1990
C2		195–215	Halls and Palmer 1990
C3		E14, 186–191	Halls and Palmer 1990

NOTES: The mean directions cited in the text for different regions of the swarm were calculated from a combination of new site data tabulated in Table A1 plus some previously published data listed above together with the appropriate reference. In each case, the same selection criteria (see footnotes [h], [j], Table A1) as for the new data set were employed in deciding whether to include a site direction from previously published data in the final mean.

TABLE A3. Tests for significant differences at the 95% confidence level between mean paleomagnetic directions from different sectors of the swarm, using the criteria established by McFadden and Lowes (1981) and McFadden and Jones (1981)

Data sets	N	H1	H2	Data sets	N	H1	H2
NW N vs. KZ N	9/18(33)	**no (no)**	**yes (yes)**	MK R vs. TM R	14/15	yes	yes
NW N vs. SE N	9/8	**no**	**yes**	MK R vs. WL R	14/3	yes	yes
SE N vs. KZ N	8/18(33)	**no (no)**	**yes (yes)**	MK R vs. RL R	14/19	yes	yes
NW R vs. KZ R	25/5(19)	**no** (yes)	**yes** (yes)	C2 R vs. TM R	5/15	**no**	**yes**
NW R vs. SE R	25/39	**no**	**yes**	C2 R vs. WL R	5/3	**no**	**yes**
SE R vs. KZ R	39/5(19)	**no** (yes)	**yes** (yes)	C2 R vs. RL R	5/19	**no**	**yes**
HP N vs. HP R	3/11	no	no	TM R vs. WL R	15/3	**no**	**yes**
HT N vs. HT R	6/10	**no**	**yes**	TM R vs. RL R	15/19	yes	yes
TM N vs. TM R	5/15	yes	yes	RL R vs. WL R	19/3	no	no
RL N vs. RL R	3/19	**no**	**yes**	HT N vs. HP N	6/3	no	no
MK N vs. MK R	15/14	no	no	HT N vs. TM N	6/5	yes	yes
OL R vs. HP R	4/11	no	no	HT N vs. RL N	6/3	**no**	**yes**
OL R vs. HT R	4/10	**no**	**yes**	HT N vs. MK N	6/15	**no**	**yes**
OL R vs. MK R	4/14	yes	yes	HT N vs. C1 N	6/11	**no**	**yes**
OL R vs. C2 R	4/5	**no**	**yes**	HT N vs. C3 N	6/5	**no**	**yes**
OL R vs. TM R	4/15	**no**	**yes**	HP N vs. TM N	3/5	**no**	**yes**
OL R vs. WL R	4/3	**no**	**yes**	HP N vs. RL N	3/3	no	no
OL R vs. RL R	4/19	**no**	**yes**	HP N vs. MK N	3/15	**no**	**yes**
HP R vs. HT R	11/10	**no**	**yes**	HP N vs. C1 N	3/11	**no**	**yes**
HP R vs. MK R	11/14	yes	yes	HP N vs. C3 N	3/5	**no**	**yes**
HP R vs. C2 R	11/5	**no**	**yes**	TM N vs. RL N	5/3	yes	no
HP R vs. TM R	11/15	**no**	**yes**	TM N vs. MK N	5/15	yes	yes
HP R vs. WL R	11/3	**no**	**yes**	TM N vs. C1 N	5/11	yes	yes
HP R vs. RL R	11/19	yes	yes	TM N vs. C3 N	5/5	yes	yes
HT R vs. MK R	10/14	yes	yes	RL N vs. MK N	3/15	**no**	**yes**
HT R vs. C2 R	10/5	**no**	**yes**	RL N vs. C1 N	3/11	no	no
HT R vs. TM R	10/15	**no**	**yes**	RL N vs. C3 N	3/5	no	no
HT R vs. WL R	10/3	**no**	**yes**	MK N vs. C1 N	15/11	yes	yes
HT R vs. RL R	10/19	**no**	**yes**	MK N vs. C3 N	15/5	no	no
MK R vs. C2 R	14/5	no	no	C1 N vs. C3 N	5/11	no	no

NOTES: N, the number of sites (≥ 3) contributing to the mean from each area respectively. The values in parentheses refer to the KZ data set when the MK data are included. H1, test whether the hypothesis that the two populations share a common precision parameter (κ) can be rejected (yes = reject hypothesis). H2, test whether the hypothesis that the two populations share a common mean can be rejected (yes = reject hypothesis). The data were adjusted for differences in geographical location before statistical comparison. H1 employs eq. 5, McFadden and Jones (1981). If the hypothesis is not rejected, eq. 29 of McFadden and Lowes (1981) can be used to distinguish the populations. Otherwise, only an approximate test using eq. 33 of McFadden and Jones (1981) can be used. The most significant results (same κ, different mean) have been highlighted in bold face. The first six data comparisons are between groups of data from northwest of (NW: HP, HT, OL), within (KZ: C1, C2, C3 $\pm$ MK), and southeast of (SE: RA, RL, TM, WL) the KSZ. Because of the less stringent site selection criteria, the MK data set has anomalously high dispersion, which prevents employing the more reliable H2 test. The rest of the data comparisons are between individual data sets from separate segments of the swarm. High-quality RA data are too sparse for them to be considered separately.